Problem 12. Prove that $\bar{A} = A \cup A'$. Deduce that A is closed in X if and only if $A' \subseteq A$.

Solution: Let $x \in \bar{A}$. Now,

$$\begin{aligned} x \in \bar{A} &\iff \text{every neighbourhood of } x \text{ intersects } A \\ &\iff x \in A \text{ or } x \in A' \\ &\iff x \in A \cup A'. \end{aligned}$$

Since x was chosen arbitrarily, we can say $\bar{A} = A \cup A'$.

For the second part, assume A is closed. Then $\bar{A} = A \implies A \cup A' = A \implies A' \subseteq A$. On the other hand, if we assume $A' \subseteq A$, then we get $\bar{A} = A \cup A' = A$. Thus, in this case, we can conclude that A is closed. As both directions of the double implication hold, we have A is closed in X if and only if $A' \subseteq A$. □

Problem 13. Prove that $\mathrm{Int}(A) \cap \mathrm{Bd}(A) = \varnothing$, and $\bar{A} = \mathrm{Int}(A) \cup \mathrm{Bd}(A)$.

Solution: We know, $\mathrm{Int}(A) \cap \mathrm{Bd}(A) = \mathrm{Int}(A) \cap \bar{A} \cap \overline{X \setminus A}$. For the sake of contradiction, assume there is a point $x \in \mathrm{Int}(A) \cap \mathrm{Bd}(A) \implies x \in \mathrm{Int}(A) \cap \bar{A} \cap \overline{X \setminus A}$. As $x \in \overline{X \setminus A}$, every neighbourhood of x must intersect $X \setminus A$. Thus, $\mathrm{Int}(A) \cap \overline{X \setminus A} \neq \varnothing$. But this is impossible as $\mathrm{Int}(A) \subseteq A$ ($\Rightarrow\!\!\!\!\!\!\Leftarrow$). Thus, $\mathrm{Int}(A) \cap \mathrm{Bd}(A) = \varnothing$.

For the second part, let $x \in \bar{A}$. Again, this happens if and only if every neighborhood of x intersects A. Now, either there exists a neighborhood U of x such that $x \in U \subseteq A$, or there is no such neighborhood. In the first case, $x \in U \subseteq A \implies x \in \mathrm{Int}(A)$ since $\mathrm{Int}(A)$ is the largest open set contained in A. In the second case, every neighborhood V of x intersects $X \setminus A \implies x \in \overline{X \setminus A} \implies x \in \mathrm{Bd}(A)$. Combining the two cases, $x \in \mathrm{Int}(A) \cup \mathrm{Bd}(A)$. Since $x \in \bar{A}$ was chosen arbitrarily, $\bar{A} \subseteq \mathrm{Int}(A) \cup \mathrm{Bd}(A)$.
For the other way inclusion, notice that $\mathrm{Int}(A) \subseteq A \subseteq \bar{A}$. Also, $\mathrm{Bd}(A) = \bar{A} \cap \overline{X \setminus A} \subseteq \bar{A}$. Therefore, $\mathrm{Int}(A) \cup \mathrm{Bd}(A) \subseteq \bar{A}$. Hence, we conclude that $\bar{A} = \mathrm{Int}(A) \cup \mathrm{Bd}(A)$. □

Problem 14. Prove that every order topology is Hausdorff.

Solution: Let $X \neq \varnothing$ and $\mathscr{T}$ be the order topology on X. Let $x_1, x_2 \in X$ such that $x_1 \neq x_2$. Consider the basic open set $(x_1, x_2) = \{x \in X : x_1 < x < x_2\} \in \mathscr{T}$. Now, there can be two cases -

Case 1. $\underline{(x_1, x_2) = \varnothing}$. Observe that

$$(-\infty, x_2) \;=\; \bigcup_{a \in X,\ a < x_2} (a, x_2) \;\in\; \mathscr{T}.$$

Similarly,

$$(x_1, \infty) \;=\; \bigcup_{b \in X,\ b > x_1} (x_1, b) \;\in\; \mathscr{T}.$$

Also, $x_1 \in (-\infty, x_2)$ and $x_2 \in (x_1, \infty)$. Crucially, $(-\infty, x_2) \cap (x_1, \infty) = \varnothing$.

Case 2. $(x_1, x_2) \neq \varnothing$. Take any point $c \in (x_1.x_2)$. As we saw earlier, $(-\infty, c)$ and (c, ∞) are open in X. Note that $x_1 \in (-\infty, c)$, $x_2 \in (c, \infty)$, and $(-\infty, c) \cap (c, \infty) = \varnothing$.

Thus, in both cases, we found two disjoint open sets in X with one containing x_1 and the other containing x_2. As $x_1, x_2 \in X$ were chosen arbitrarily, we can say that $(X, \mathscr{T})$ is Hausdorff. Since $(X, \mathscr{T})$ was also chosen arbitrarily, we can say that every order topology is Hausdorff. □

Problem 15. Let X be a T_1 space. Prove that for each pair x_1, x_2 of distinct points, there exist open sets U_1 and U_2 containing x_1 and x_2, respectively, such that$x_1 \notin U_2$ and $x_2 \notin U_1$.

Solution: Let x_1, x_2 be a pair of distinct points in X. X being a T_1 space, singletons $\{x_1\}$ and $\{x_2\}$ are closed in X. Therefore, $X \setminus \{x_1\}$ and $X \setminus \{x_2\}$ are open in X. Now, $x_2 \in X \setminus \{x_1\}$ (because $x_1 \neq x_2$) but $x_2 \notin X \setminus \{x_2\}$. Similarly, $x_1 \in X \setminus \{x_2\}$ (because $x_1 \neq x_2$) but $x_1 \notin X \setminus \{x_1\}$. As $x_1, x_2 \in X$ were chosen arbitrarily, we can say that for each pair x_1, x_2 of distinct points, there exist open sets U_1 and U_2 containing x_1 and x_2, respectively, such that$x_1 \notin U_2$ and $x_2 \notin U_1$. □